

Research Proposal

A MLLM Agent Framework for Individual Mobility Prediction

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Abstract

This study aims to construct an MLLM agent framework for individual mobility simulation, powered by the advanced Multimodal Large Language Model (MLLM), GPT-4.1. The framework deeply integrates fine-grained urban data, including POIs, road networks, demographic profiles, and individualized behavioral preferences. The model is capable of simulating individuals' daily activity sequences and next point-of-interest (POI) choices. This approach transcends traditional function-based prediction models by enabling agents to see and perceive the urban environment for the first time, thereby making environmentally-aware decisions.

1 Problem Background

Individual mobility prediction is central to urban traffic management, smart city construction, and the provision of personalized services. However, traditional models have long had significant limitations. Existing methods mainly rely on single-modal data such as historical trajectory records or POI attribute statistics. These models first lack the ability to "perceive" the urban environment visually and cannot interpret visual factors such as streetscapes and architectural styles—factors that have an important impact on human mobility choices. Traditional individual mobility prediction models lack the support of a multi-agent architecture and often use a single model to complete all tasks, making it difficult to achieve specialized division of labor and collaboration.

2 Research Objectives

- **To construct a multimodal mobility simulation framework** , capable of generating detailed daily activity sequences.
- **To achieve accurate predictions for both full-day activity sequences and immediate next POI choices**, ensuring temporal and logical consistency through a constrained generation process.
- **To develop a multi-agent architecture** where specialized LLM agents collaborate to perform individual profiling, activity planning, and location selection.

3 Research Methods

3.1 Data Preparation

Input Data:

1. Urban Built Environment :

- **POI Data:** Includes POI categories, locations, and **dynamic status information**.
- **Visual Data:** High-resolution imagery of various urban areas to provide visual context of the built environment.
- **Network Data:** A topological road network, including pedestrian and cycling paths, for calculating travel times.

2. Human Attribute Data:

- **Static Profile:** Gender, age, income, occupation.
- **Dynamic Profile:** Current location, historical activity sequences, and preference tags.

Output:

1. **Individual Activity Sequence:** A detailed activity chain for the next 24 hours, including time, location, and activity type.
2. **Next POI Prediction:** The specific next point-of-interest immediately following the current activity in the sequence.

3.2 Model Architecture

We will select **GPT-4.1** as the core MLLM to support the framework’s required multimodal understanding and complex reasoning capabilities. The entire architecture will be a **collaborative multi-agent framework**.

1. **Agent 1: Persona Profiler** This agent is responsible for creating a comprehensive and structured profile for each individual.

- **Input:** The individual’s raw static and dynamic attributes.
- **Function:** Processes the input to generate a structured "Persona" profile. For example: "User A: A young white-collar worker who commutes by subway. Weekend activities are primarily leisure-oriented. Frequently visited POI types include cafes and parks. Behavioral patterns show a gym visit every Friday at 18:00." This profile will serve as the foundational context for all subsequent agents.

2. **Agent 2: Activity Sequence Planner** This agent acts as a strategic planner, outlining the day’s macroscopic activities.

- **Input:** The "Persona" profile from Agent 1 and the current context.
- **Function:** Generates a logically coherent, high-level activity sequence for the remainder of the day.

3. **Agent 3: Multimodal Location Selector** This agent is the tactical decision-maker, responsible for grounding the planned activities at specific real-world locations.

- **Input:** A specific activity instruction from Agent 2 (e.g., "go to a supermarket"), the "Persona" profile, and the agent’s current location.
- **Function:** Selects the final POI through a multimodal reasoning process:
 - **Candidate Retrieval:** First, retrieves a list of candidate POIs based on functional requirements (e.g., supermarkets within a 15-minute travel radius).
 - **Multimodal Filtering and Reasoning:** The LLM then receives a consolidated view of each candidate, including:
 - * **Structured Data:** Ratings, real-time crowdedness.
 - * **Visual Context:** Satellite imagery of the POI’s surrounding area.

- * **Persona Context:** The individual’s preferences.
- The agent reasons across these inputs to make a final choice. For example:
 ”Candidate A is closer, but real-time data shows it is very crowded. Candidate B is slightly farther but, as inferred from imagery, is in a less dense area and is larger. Given the persona’s preference to avoid crowds, Candidate B is the optimal choice.”

4 Main Required Resources

Resource Type	Details
Datasets	Deep data for a single city: Official POI databases (e.g., NYC Open Data), user-generated content platforms (for photos and reviews, e.g., Google Maps).
Computational	GPUs for model training/inference (requirements are lower if using APIs).

5 Timeline

- **Week 1:** Collection, cleaning, and initial integration of multimodal data.
- **Week 2:** Feature engineering and splitting of datasets into training/validation/test sets; writing of literature review and introduction.
- **Week 3:** Implementation of Agent 1.
- **Week 4:** Implementation of Agent 2.
- **Week 5:** Implementation of Agent 3.
- **Week 6:** Integration of the three agents; writing of the Methods section.
- **Week 7:** Large-scale simulation experiments.
- **Week 8:** Comparison with baseline models.
- **Week 9:** Supplementary sensitivity analysis experiments; writing of the experimental section of the paper.
- **Week 10:** Writing of the Results and Discussion sections and optimization of all figures.

- **Week 11:** Completion of the first full draft of the paper; internal review and revision.
- **Week 12:** Organization of the codebase and submission of the paper.